

9708/32/o/n/23

1-C:

Marginal utility is considered the change in total utility when an additional unit is consumed.

To calculate marginal utility of the 4th bar, the formula will be: $TU \text{ at 4 bars} - TU \text{ at 3 bars}$.

For Consumer Y:-

Total utility at 4 bars = 70

Total utility at 3 bars = 65

Marginal utility = $70 - 65 \rightarrow 5$

For Consumer Z:-

Total utility at 4 bars = 40

Total utility at 3 bars = 45

Marginal utility = $40 - 45 \rightarrow -5$ (**marginal utility is negative**)

The difference in marginal utility between Consumer Y and Consumer Z is:

$+5 - (-5)$

• $= 10$

This explains why key C is correct.

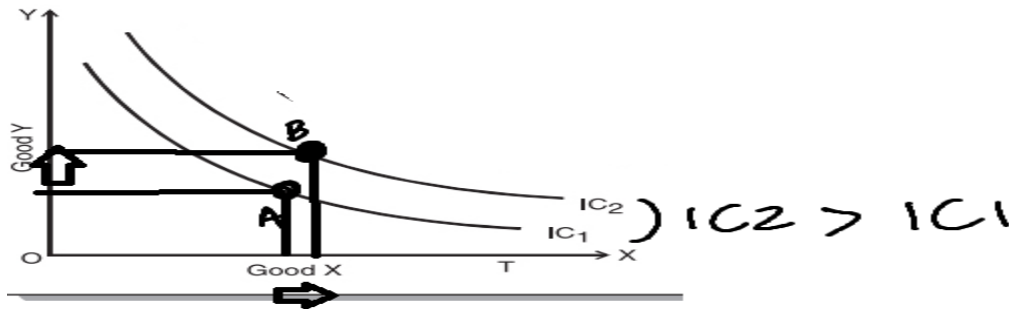
2-C:

C) The curves are typically downward sloping, convex, and never intersect or cross each other.

Indifference curves do not intersect because this would suggest that a consumer is indifferent between two bundles of goods, one of which contains more of both goods than the other.

This is because, the further the indifference curve is from the origin the greater will be the satisfaction that it will yield. This further explains why key C will be correct.

- Imagine two indifference curves, IC1 and IC2,
- Since IC2 is above IC1, combination B must contain more of both goods than combination A.
- This means that combination B is unambiguously better than combination A, since it contains more of both goods.
- **Therefore, the consumer cannot be indifferent between the two combinations, and the curves cannot cross.**



A) A budget line shows the combination of goods that a consumer can purchase with the given income.

B) An indifference curve is not a straight line; instead, it is convex to the origin. This is because, it represents decreasing opportunity cost because of the law of diminishing marginal utility in action.

C) The gap between indifference curves is not always constant.

3-D:

-> **Negative externalities**, such as air pollution from car emissions, can lead to market failure because the free-market does not consider the external costs. This means that the market outcome is not efficient, as the marginal social cost (MSC) exceeds the marginal private cost (MPC), representing the existence of marginal external cost (MEC)

To address this market failure, government intervention is likely necessary to internalize the external costs and achieve economic efficiency. Possible interventions include: Imposing a tax on car emissions (e.g., a carbon tax). **This explains why key D is correct.**

Here's why the other options are incorrect:

A) Car manufacturers will not automatically bear the external costs; government intervention is needed to internalize these costs.

B) Market failure occurs when Marginal social benefits are not equal to Marginal social costs; simply equating private and external costs does not resolve the issue.

C) While some people may choose to use buses, this is not a sufficient response to address the market failure; government intervention is needed to achieve economic efficiency.

4-B:

A) Monopolistic competition is a market structure characterized by many firms producing differentiated products, with free entry and exit. In this scenario, the merger has resulted in a single firm with 80% market share, which doesn't fit the characteristics of monopolistic competition.

There are still thousands of independent taxi drivers, but the dominant firm has a significant market share, making it different from monopolistic competition.

B) Oligopoly is a market structure with a few large firms competing with each other.

-> Before the merger, the market was likely an oligopoly with two large firms and many independent taxi drivers.

-> After the merger, one firm has gained 80% of the market share, making it a dominant player, and the market has essentially become a monopoly.

C) Perfect competition is a market structure with many firms producing homogeneous products, with free entry and exit, and no single firm having market power. The scenario doesn't suggest that the market was ever in a state of perfect competition, as there were already two large firms and many independent taxi drivers.

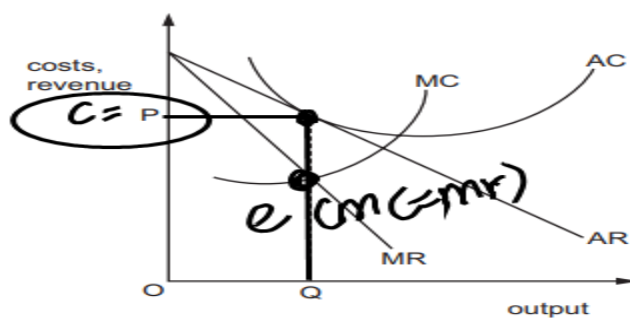
-> After the merger, the market has become more concentrated, but it hasn't changed from perfect competition to oligopoly, as it was never in a state of perfect competition to begin with.

D) While the market was indeed an oligopoly before the merger, the significant concentration of market share and the emergence of a dominant player mean that the market structure has changed.

-> The market is no longer the same oligopoly as before, as one firm now has a significant market share, making it closer to a monopoly.

5-D:

- A- Since the profit maximizing firm is operating where $MC=MR$, and not where $MC=AR$, it is not allocatively efficient.
- B- It is not making losses. A loss would have been made when price (AR) would have been less than ATC.
- C- It is not making abnormal profits. Supernormal profits would have been made when price (AR) was greater than ATC.
- D- Yes, this is correct because at the profit maximizing equilibrium where $MC=MR$, its price (AR) is equal to ATC. This implies that a firm is making **normal profits or zero-economic profits**.



6-D:

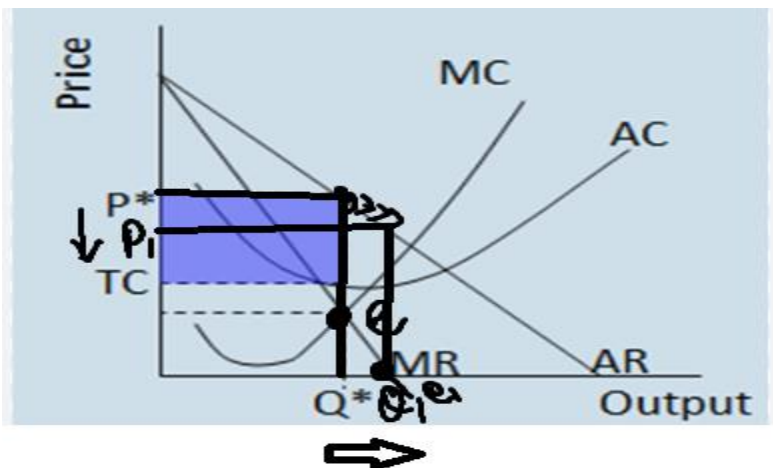
When a firm produces where $MC=MR$, it is maximizing its profits. By changing its aim to produce where $MR=0$, the firm is no longer maximizing profits. Instead, it is sacrificing profits to achieve **the objective of sales revenue maximization**. Since this does not mean that the firm is wishing to gain more profit, option D becomes correct.

A) The firm wishes to increase sales: True, producing where $MR=0$ means the firm is willing to sell more units, even if it means reducing price.

B) The firm wishes to increase its output: True, producing where $MR=0$ means the firm is willing to produce more, even if it means reducing price.

C) The firm wishes to reduce prices: True, producing where $MR=0$ means the firm is willing to reduce price to increase sales and output.

This can be seen in the graph below.



7-D:

Third-degree price discrimination occurs when a firm charges different prices to different groups of consumers based on their **price-elasticity of demand**. This is possible when the firm can segment the market into distinct groups with different price elasticities of demand.

Condition D) The firm is able to keep the different market segments separate is essential for third-degree price discrimination to work. This means that the firm must be able to prevent resale between segments, ensuring that each group pays the price intended for them.

A) If consumers are able to buy cheaply and then sell the product at a higher price, this would undermine the firm's price discrimination strategy. One important condition that enables price-discrimination is that a firm should be able to prevent resale between segments, ensuring that each group pays the price intended for them.

B) Price elasticity of demand should not be the same in each market segment, or else, price-discrimination would not be possible.

C) If the firm faces a perfectly elastic demand curve, this would mean that the firm has no market power, and price discrimination would not be possible.

8-A:

Moral hazard refers to the tendency of people to take greater risks or engage in riskier behavior when they are insulated from the consequences of their actions.

A- In this case, people with a higher risk of unemployment are more likely to undertake training to improve their skills and reduce their risk of unemployment. This is a responsible behavior, not a moral hazard.

B) People with free dental care may be less likely to take care of their teeth because they don't bear the cost of dental care.

C) People with insurance against car theft may be more likely to park their car in high-crime areas because they don't bear the full cost of theft.

D) People with unemployment benefits may be less likely to undertake retraining because they receive benefits regardless of their efforts to find new employment.

This represents that all other options represent “moral hazard”

9-A:

Marginal Private Cost (MPC) Curve:

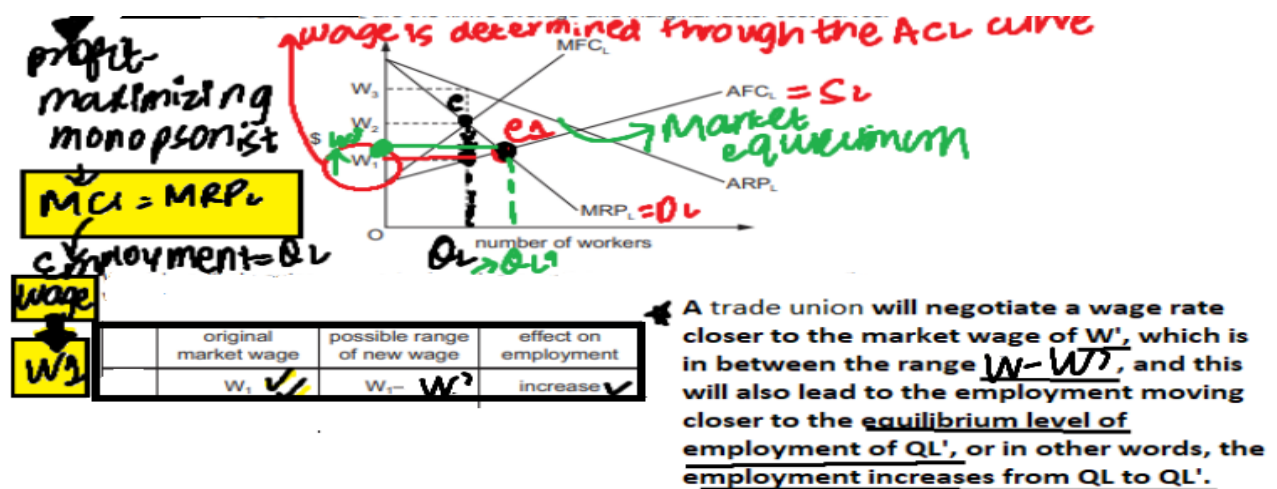
- The MPC curve represents the private cost borne by the airline, such as fuel costs, labor costs, and maintenance costs.
- When fuel costs decline, the MPC curve shifts downward, indicating that the airline's cost of operating an additional aircraft has decreased.
- The downward shift of the MPC curve shows that the airline's marginal cost of production has decreased.

Marginal External Cost (MEC) Curve:

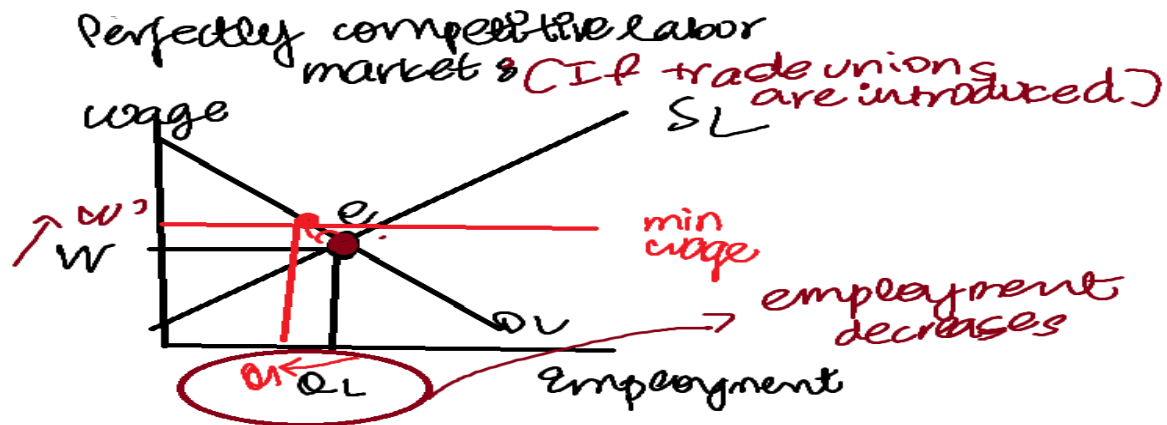
- The MEC curve represents the cost borne by the third party who is not directly involved in the transaction, such as:-noise pollution, air pollution, and congestion.
- When aircraft design improves, leading to quieter aircraft, the MEC curve shifts downward, indicating that the marginal external cost of air travel in form of noise pollution has decreased.

10-B: Effect on Firm L (monopsony employer):

When trade unions are introduced in an imperfect labor market where there is an existence of a monopsony, it is most likely that alongside an increase in wages, employment will also increase.



Firm K(perfectly competitive market):

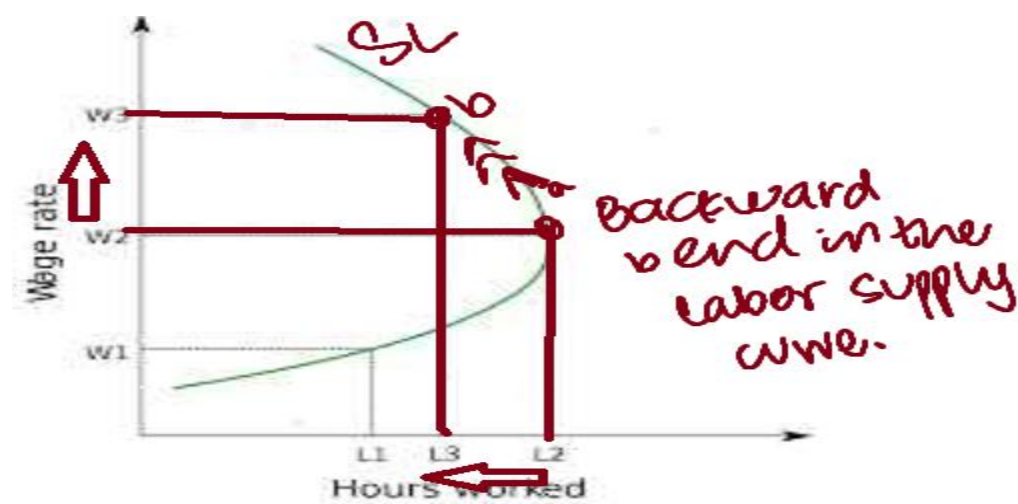


In a perfectly competitive labor market, equilibrium is where demand for labor is equal to supply for labor. When wages are raised to a level above that of the equilibrium, quantity demanded for labor that represents employment will contract, which represents that there will be an increase in unemployment.

11-C:

- **Substitution effect:** When wages rise, workers tend to substitute leisure time with work hours, as working becomes more lucrative. This effect increases the supply of labor.
- **Income effect:** However, as wages rise, workers' incomes also increase, enabling them to meet their target income with fewer working hours. This effect decreases the supply of labor.

In a backward-sloping supply curve, the income effect dominates the substitution effect. This means that the increase in income leads workers to choose more leisure time and fewer working hours, despite the higher wage.



12-D:

A Universal Basic Income has two main economic results:

1. **Reduces the incentive to work:** A UBI provides a guaranteed minimum income, which can lead to a decrease in the labor supply as people may choose to work less, since their basic needs are already met.
2. **Improves equality:** A UBI aims to reduce poverty and income inequality by providing a safety net for all citizens, ensuring everyone has a minimum standard of living.

13-B:

- **Increasing top rates of income tax:** This makes the tax system more **progressive**, as the wealthy are taxed at a higher rate, which represents that a greater proportion of the incomes of the high income groups is taken up in taxes. The funds generated can then be used to provide means tested benefits to the lower income groups.
- **Reducing indirect taxes:** Indirect taxes, such as sales taxes or VAT are **regressive** in nature, which means that they disproportionately affect lower-income households by taking up a higher proportion of the incomes of the low income groups. Reducing them would decrease the tax burden on these households.

-> Lower indirect taxes lead to lower prices for essential items. Since low-income individuals spend a larger portion of their income on essential items, they benefit from the price reduction. The quantity demanded for essential items by low-income individuals' increases, but by a smaller percentage due to the inelastic nature of demand. As a result, there is a decrease in total expenditure. This means that low-income individuals spend less on essential items despite the increase in quantity demanded, indicating a fall in total expenditure.

- **Value of state benefits(increase):**

When the value of state benefits increases, it means that the government is providing more financial support to eligible individuals through means-tested benefits. Means-tested benefits are designed to help those who need it most, typically low-income households.

This explains why key B becomes correct.

14-B:

- Initially, area “2” was not included and no congestion charge was charged, so that means that there was a greater number of journeys that were taking place from “X to Y”
- But now, since congestion charges are extended to include area “2”, which includes the region “Y”, this means that now journeys occurring from “X” to “Y” will reduce actually.
- As a result of which, key B, becomes correct.

A – Area “1” that includes the destination “Z” already had the charge, which means that there would not be any difference in the number of journeys.

C – Again , journeys from “Y” to “Z” would not reduce because area 1 already had the charge.

D – Journeys from point Z to point Y would not reduce because the congestion charge is of the same amount, so there will be no difference.

15-A:

A – A fall in interest rate on savings accounts will reduce the incentives to save, and when because of a higher opportunity cost of spending, savings reduce and subsequently consumption increases, there will be a subsequent increase in Aggregate demand and real national income. This is why, key A will be correct.

B – When because of a fall in the prices of imported goods, import spending rises, consumption expenditure on domestically produced goods will fall. This will be likely to reduce Aggregate demand and real national income.

C – A decrease in government spending which is an autonomous injection into the circular flow will reduce Aggregate demand, and subsequently real national income will fall.

D – An increase in direct taxes will reduce the disposable income, as a result of which, consumption expenditure will fall. This will translate into a fall in Aggregate demand and real national income.

16-B:

Occupational mobility refers to the ability of individuals to switch from one job to another.

Increased government spending on education and training improves occupational mobility by: Providing individuals with new skills and qualifications, which makes them more versatile and multi-skilled. This enhances their occupational mobility, which means that they can easily and flexibly switch in between occupations especially when the structure of the economy is changing.

- A: Decreased income tax rates may increase disposable income but they don't directly improve skills or employability.
- C: Increased government spending on infrastructure may create jobs but doesn't necessarily enhance skills.
- D: Increased subsidies to firms to employ school leavers may help young people get jobs but doesn't necessarily improve their skills or occupational mobility.

17-B:

The formula for **four sector multiplier** is as follows:-

$$\text{Multiplier} = 1 / (\text{MPS} + \text{MPM} + \text{MRT})$$

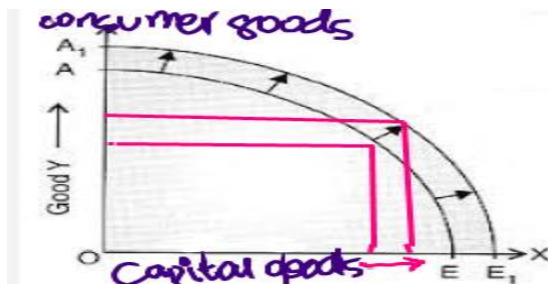
$$\begin{aligned}\text{Multiplier} &= 1 / (0.2 + 0.1 + 0.2) \\ &= 1 / 0.5 \\ &= 2\end{aligned}$$

So, the correct answer is indeed B: 2.

18-C:

A) When the productive capacity of the economy increases over a long period of time, this means that future generations will also be well-off.

B) This is incorrect because an increase in the productive capacity of the economy implies that there has been a rightward shift of the production possibility curve. This means that the output of both goods can increase.



C) When the productive capacity of a free enterprise economy increases over a long period of time, this implies that the economy can produce more goods and services. Since it can be seen in the diagram above that consumers have access to a wider range of consumer goods, this signifies an increase in choices, which represents rising standard of living. **As a result of this, option C becomes correct.**

D) Rising levels of output can lead to increased per capita incomes, but equalization is not a direct result of increased productive capacity.

19-B:

The accelerator principle suggests that investment (I) is linked to changes in GDP (Y). Specifically, it states that investment is a function of the change in GDP, rather than the level of GDP. This means that the accelerator co-efficient can be calculated using:

Change in investment/Change in Income.

The Accelerator Principle suggests that when the change in GDP grows, investment increases, and when the change in GDP declines, investment decreases. This relationship is based on the idea that businesses invest in new capital goods and technology to meet increased demand. **This explains why key B is correct.**

20-A:

An increase in the budget deficit means the government is spending more than it's taking in through taxes. This increased government spending injects more money into the economy, boosting aggregate demand, also known as **effective demand**. This explains why key A is correct.

- B: Because of a rise in Aggregate demand, production and demand for labor will increase, which results in a decrease in demand-deficient unemployment.
- C: An increase in budget deficit will increase Aggregate demand and when Aggregate demand outpaces Aggregate supply, there is likely to be a shortage, which results in an increase in demand pull inflation.
- D: An increase in budget deficit represents an expansionary fiscal policy, which will result in an increase in Aggregate demand, because of which, real incomes will rise. This will subsequently bring about an increase in volume of spending.

21-C:

The table gives details of the inflation and unemployment rates for different countries in 2008 and 2011.

country	inflation rate (% per annum)		unemployment rate (% of workforce)	
	2008	2011	2008	2011
Canada	2.38	2.89	6.15	7.47
France	3.16	2.29	7.81	9.68
Germany	2.75	2.48	7.60	5.98
Japan	1.37	-0.28	3.99	4.55

The question is asking about the historical phenomenon of the "Phillips Curve", which represents a **negative relationship between inflation and unemployment**.

The Phillips Curve shows that, there is a trade-off between these two variables: when unemployment was low, inflation was high, and when unemployment was high, inflation was low. This meant that policymakers could potentially "trade off" one for the other, by using monetary or fiscal policy to reduce unemployment at the cost of higher inflation, or to reduce inflation at the cost of higher unemployment.

It can be seen that a **negative relationship between inflation and unemployment** is only observed in France and Japan, which is why, key C becomes correct.

22-C:

Option A: Creating a more **equal distribution of income and wealth** can have a positive impact on economic development. By redistributing income and wealth, governments can reduce poverty, and increase access to education and healthcare. This can further drive economic development.

Option B: Diverting government spending from industry to education can also promote economic development. **Education is a key driver of economic development**, as it increases human capital and leads to a more educated and skilled workforce.

Option C: Encouraging the extraction of finite natural resources, such as oil or gas, can boost economic growth in the short term but may not promote economic development. This is because, it can also lead to **depletion of natural resources, and subsequently decreased economic activity in the long term**.

Option D: Environmental regulations can lead to increased investment in clean technologies, which can drive innovation and economic growth. **Additionally, reducing air pollution can lead to improved public health, which can increase productivity and reduce healthcare costs. This will drive economic development.**

	Economic Growth	Economic Development
Definition	Increase in economic activity, measured by GDP	Improvement in living standards, social welfare, and economic structure
Focus	Quantitative expansion	Qualitative transformation

23-A:

The Laffer curve suggests that as tax rates increase, government revenue initially increases, but eventually decreases as high tax rates discourage economic activity. The curve suggests that there is an optimal tax rate that maximizes government revenue.

Cuts in direct taxes, such as income tax, are a policy measure that might be based on the concept of the Laffer curve. The idea is that reducing tax rates will encourage economic activity, leading to increased tax revenue for the government.

This explains why key A will be correct.

24-B:

When the exchange rate appreciates , it makes:

1. Imports cheaper: When import prices fall, the quantity demanded may increase, but by a smaller percentage if the demand is inelastic. This occurs in a situation where raw materials are imported. This means that the total expenditure on imports may actually decrease, even though the quantity imported increases. When because of this, costs of production fall, Aggregate supply will subsequently increase. This will subsequently lead to a fall in **imported inflation**.

2. Exports more expensive: When exported goods become relatively expensive in terms of foreign currency, quantity demanded of exported goods may contract and that too by a relatively higher percentage when the demand for them is elastic. This will subsequently result in a fall in export revenues and when autonomous injections into the circular flow decrease, Aggregate demand will fall. This translates into a decrease in production and decrease in demand for labor, which results in an **increase in demand-deficient unemployment**.

25-A:

1. **Low saving ratio:** People in low-income countries tend to have limited disposable income, which means they have less money to save. Additionally, many low-income countries lack developed financial systems, making it harder for people to save and invest.
2. **Low capital-output ratio:** This means that a small amount of capital (e.g., factories, machinery) is used. This can be due to various factors like a lack of investment. This explains why key A will be correct.

26-B:

Over the last 25 years, global economic growth has been accompanied by:

- Increased international trade: Globalization has led to a significant increase in international trade, with countries becoming more interconnected and interdependent.(B)
- Growing urbanization: Many people have moved from rural areas to cities, leading to rapid urbanization and changes in economic and social structures.(C)
- Increased atmospheric pollution: Economic growth has often been linked to increased energy consumption and pollution, contributing to environmental degradation.(D)
- Depletion of non-renewable resources: The increased economic activity has led to a greater consumption of natural resources, including non-renewable resources like fossil fuels.(A)

International trade has actually increased significantly over the past 25 years, with the rise of globalization. This explains why key B will be correct, **because decreased international trade has not accompanied global economic growth over the last 25 years.**

27-B:

The correct answer is B: It has moved closer to zero.

The Gini coefficient is a measure of income inequality, ranging from 0 (perfect equality) to 1 (perfect inequality).

- A Gini coefficient of 0 represents perfect equality, where everyone has the same income.
- A Gini coefficient of 1 represents perfect inequality, where one person has all the income.

When the Gini coefficient moves closer to zero, it indicates that the income distribution is becoming more equal, with fewer disparities between the rich and the poor.

- A: Moving closer to 1 would indicate increased inequality.
- C and D: The Gini coefficient is always a positive value, so it cannot move from negative to positive or vice versa.

28-C: The Suez Canal incident highlights the interdependence arising from global trade. The blockage of the canal:

- Disrupted supplies, causing shortages in markets worldwide
- Affected over 350 cargo ships, demonstrating the vast network of global trade

This event showcases the increased interdependence between countries, which is a key consequence of globalization. The world's economies are now more connected than ever, **and disruptions in one region can have far-reaching effects globally.**

- A: While globalization can lead to lower prices, this event highlights the potential costs of disruptions.
- B: While globalization increases choices for consumers, this event focuses on supply chain disruptions.
- D: Increased migration of labor between countries is not related to this event.

29-C:

Economic growth refers to an increase in real GDP per capita. **In Spain, real GDP per capita increased from \$29,399 to \$32,283, indicating economic growth.** In Namibia, real GDP per capita decreased from \$6,113 to \$6,013, indicating a decline in economic growth.

Economic development refers to improvements in living standards, health, education, and overall well-being. Health spending per capita is a proxy for economic development. **In Spain, health spending per capita decreased from \$2,671 to \$2,506, indicating a decline in economic development. In Namibia, health spending per capita decreased from \$452 to \$447, also indicating a decline in economic development.**

Therefore, only Spain experienced economic growth, while neither country experienced economic development.

30-C:

The diagram below explains that only education and life expectancy have comparable changes to the overall change in HDI, which makes key C correct.

